An Internship Report

Of

DATA SCIENCE MASTER

Submitted

To

School of Computer Science and Engineering
IILM University

Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Name of Student: SIDRA FATIMA

ROLL NO:2410031342

Batch: [2026-27]

Internship Duration: June – August 2026

AICTE – EduSkills Virtual Internship Program

CANDIDATE'S DECLARATION

I, SIDRA FATIMA, do hereby declare that the internship report titled “Data Science Master” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report is based on the learning and activities associated with the 8-weeks AICTE – EduSkills Virtual Internship Program completed during June – August 2026. All references used in preparing this report have been acknowledged appropriately. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement, and I shall be responsible for any copyright violation in this regard.

Signature:-

Student Name: SIDRA FATIMA

Student ID: STU69e6083525c761776683061

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their guidance, constructive criticism and advice during this work. I am sincerely grateful to everyone who contributed to my learning experience.

I would like to thank the AICTE – EduSkills Virtual Internship Program team and EduSkills Academy for providing me with the opportunity to participate in the Data Science Master Virtual Internship. The program provided a structured platform to strengthen my understanding of data science concepts and to develop practical, analytical and problem-solving skills. I am also thankful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I also express my gratitude to my parents and everyone who encouraged me throughout the internship.

Signature:

Student Name: SIDRA FATIMA

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3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF
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अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills®
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Sidra Fatima

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

G.V. Pai

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4820a932c55cd3b60e7e

Student ID: STU69e6083525c761776683061

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

4.1 Introduction

This report presents a detailed account of the 8-weeks Data Science Master Virtual Internship undertaken through the AICTE – EduSkills Virtual Internship Program. The internship was completed during June – August 2026 and the completion certificate identifies the domain as "Data Science Master". The internship provided an opportunity to strengthen practical understanding of the data science lifecycle, from collecting and preparing data to analysing patterns, building predictive models and communicating results.

The central focus of data science is to transform raw data into useful information and actionable insights. During a structured learning program, concepts such as data preprocessing, exploratory data analysis, statistical reasoning, data visualization, machine learning and model evaluation can be connected into a complete workflow. This report presents the domain, objectives, scope, workflow, methodology and expected learning outcomes in the same report structure as the provided internship-report template.

The certificate confirms successful completion of the 8-weeks virtual internship during June – August 2026 and records the internship domain as Data Science Master.

4.2 Organization Profile

EduSkills is an organization working under the theme "Nation Building Through Skills" and conducts industry-oriented virtual internship programs in association with the All India Council for Technical Education (AICTE). The AICTE – EduSkills Virtual Internship Program provides students with a structured environment to learn emerging technologies and gain practical exposure through guided learning activities and assessments.

The internship certificate provided for this report carries the AICTE, National Internship Portal and EduSkills branding and states that the Data Science Master Virtual Internship was completed during June – August 2026. Siemens is shown as the supporting organization on the certificate. The program therefore serves as a bridge between academic learning and practical technology-oriented skill development.

4.3 Problem Statement

Modern organizations generate large volumes of structured and unstructured data, but raw data cannot directly support reliable decision-making. Data may contain missing values, duplicate records, inconsistent formats, outliers and irrelevant variables. Without a systematic process for cleaning, exploring and modelling data, useful patterns can remain hidden and analytical conclusions may be unreliable.

The problem addressed by a Data Science Master internship is therefore to understand and apply a complete data science workflow: obtaining data, cleaning and transforming it, exploring relationships, visualizing important patterns, selecting appropriate analytical or machine-learning techniques, evaluating results and communicating insights clearly.

4.4 Project Objectives

- To understand the end-to-end data science lifecycle and the role of each stage in an analytical project.
- To strengthen Python-based data analysis and problem-solving skills.
- To learn data cleaning, preprocessing and feature preparation techniques.
- To perform exploratory data analysis and identify meaningful patterns and relationships.
- To create effective visualizations for communicating data-driven findings.
- To understand fundamental machine-learning concepts and the model-development workflow.
- To evaluate analytical or predictive models using suitable performance measures.
- To develop the ability to interpret results and communicate conclusions in a clear and structured manner.
- To connect theoretical concepts with practical data-oriented tasks through a structured virtual internship.

4.5 Scope of the Project

The scope of the Data Science Master domain covers the major stages required to convert raw data into useful analytical outcomes. It includes data understanding, data preparation, exploratory analysis, visualization, basic statistical interpretation, machine-learning workflows and model evaluation. The scope also includes presenting findings in a way that can be understood by non-technical users.

The internship was conducted in a virtual learning environment. The certificate confirms the internship domain and duration, while the detailed technical workflow described in this report is a report-format representation of the Data Science Master learning domain rather than a claim about a specific live production system.

4.6 Technologies and Tools Used

Area	Tools / Technologies
Programming & Analysis	Python
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Development Environment	Jupyter Notebook / Google Colab
Machine Learning	Scikit-learn
Data Storage / Input	CSV, Excel and other structured datasets
Model Evaluation	Accuracy, Precision, Recall, F1-score, MAE/MSE/RMSE as appropriate

4.7 Data Science Workflow / System Architecture

Data Source / Dataset

↓

Data Cleaning & Preprocessing

↓

Exploratory Data Analysis (EDA)

↓

Feature Engineering / Selection

↓

Model Building / Statistical Analysis

↓

Model Evaluation & Validation

↓

Visualization, Interpretation & Reporting

The workflow begins with a dataset and proceeds through preparation and exploration before analytical or machine-learning modelling. The output is evaluated and then communicated using visualizations and a structured report. This layered workflow helps maintain consistency and reduces errors caused by directly modelling unprocessed data.

4.8 Methodology

The internship methodology is organized as a progressive learning workflow. Each stage builds on the previous stage so that the learner can understand not only individual techniques but also how they fit into a complete data science project.

Week	Module	Description
1	Data Science Foundations	Introduction to the data science lifecycle, data types, analytical thinking and the role of Python in data analysis.
2	Python for Data Analysis	Working with Python, NumPy and Pandas for loading, inspecting, transforming and manipulating datasets.
3	Data Cleaning & Preprocessing	Handling missing values, duplicates, inconsistent data, outliers and preparing variables for analysis.
4	Exploratory Data Analysis	Using summary statistics and exploratory techniques to identify trends, distributions, relationships and anomalies.
5	Data Visualization	Creating meaningful charts and plots to communicate patterns and analytical findings effectively.
6	Machine Learning Fundamentals	Understanding supervised and unsupervised learning concepts, feature preparation, training and validation.
7	Model Evaluation & Interpretation	Evaluating models with appropriate metrics, comparing results and interpreting analytical outcomes.
8	Applied Data Science Project	Combining the learned workflow into an end-to-end data science exercise and documenting the results.

A typical task begins by understanding the objective and dataset. Data is then cleaned and transformed so that the analysis is reliable. Exploratory analysis is used to discover patterns, followed by visualization to make those patterns easier to interpret. Where prediction is required, an appropriate machine-learning model is trained and evaluated. Finally, the findings are documented and communicated.

4.9 Expected Outcomes

- Improved understanding of the complete data science lifecycle.
- Ability to load, clean, transform and analyse datasets using Python-based tools.
- Ability to conduct exploratory data analysis and identify important patterns.
- Improved skill in creating and interpreting data visualizations.
- Understanding of fundamental machine-learning workflows and model evaluation.
- Ability to select suitable evaluation metrics according to the analytical problem.
- Improved analytical thinking, problem-solving and data interpretation skills.
- Ability to present data-driven findings in a structured report.
- Greater confidence in applying data science concepts to practical datasets.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued through the AICTE – EduSkills Virtual Internship Program is attached in Chapter 3 of this report. The certificate identifies Sidra Fatima as the recipient and records successful completion of the 8-weeks Data Science Master Virtual Internship during June – August 2026. It also displays the Student ID and Certificate ID and shows Siemens as the supporting organization.

The certificate image included in this report has been reproduced from the certificate supplied with the internship-report request. No separate offer letter or communication proof was supplied with the current files, so no additional document is represented as attached.

5. BIBLIOGRAPHY/REFERENCES

1. AICTE – EduSkills Virtual Internship Program, “Certificate of Virtual Internship – Data Science Master,” June – August 2026.

2. EduSkills Academy, AICTE – EduSkills Virtual Internship Program materials, as applicable to the Data Science Master domain.

3. Python Documentation, Python programming language documentation.

4. Pandas Documentation, documentation for data manipulation and analysis.

5. NumPy Documentation, documentation for numerical computing with Python.

6. Matplotlib Documentation, documentation for data visualization.

7. Scikit-learn Documentation, documentation for machine learning in Python.

Note: The certificate supplied with this report establishes the internship title/domain, duration, recipient, institution and supporting organization. The detailed tool list and weekly methodology in this report are a structured academic report draft for the Data Science Master domain and should be adjusted if your actual internship training modules or assignments differ.

— END OF REPORT —